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Fine Tuning of the HOMO–LUMO Gap of 6-(Thiophen-2-yl) indolizino[3,2-*c*]quinolines and their Self-Assembly to Form Fluorescent Organic Nanoparticles: Rational Design and Theoretical Calculations

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In this study, we designed and synthesized novel 6-(thiophen-2-yl) indolizino[3,2-*c*]quinolines (IQs) guided by density functional theory (DFT) calculations and explored the optical properties of the IQ derivatives together with fractional atomic orbital contribution (FAOC) analysis. In accordance with the DFT predictions, the designed compound, **IQ-C9**, with a methoxycarbonyl (MC) group attached to the C9 position of the IQ scaffold, displayed improved fluorescence quantum yields (QY) in solution and in the solid state along with bathochromic shifts compared to previous analogues. TEM images demonstrated that **IQ-C9** forms well-defined nanoparticle structures (dimensions: 20×50 nm, aspect ratio: 2.3) in aqueous solution. Confocal images of **IQ-C9**-treated MCF7 cells revealed colocalization of **IQ-C9** with lipid droplets, highly dynamic subunits within cells involved in various metabolic disorders and cancers. Our results demonstrated that the analyses based on the FAOC coefficients and molecular orbital (MO) energy levels can be used as a computational method to predict spectral shifts for fluorophores and to design novel fluorescent probes for tracking dynamic cellular processes.

1. Introduction

Fused heterocyclic ring systems have gained enormous attention in various fields, including polymer chemistry, material science, and medicinal chemistry.[1–6] In recent years, numerous studies have mainly focused on the application of such systems as organic fluorophores, since fused aromatic compounds have extensive conjugated π-systems and rigid planar structures, which are necessary conditions for fluorescence. Nevertheless, most fluorescent heterocyclic scaffolds usually require extensive optimization to obtain appropriate substituents to exhibit the desired properties for practical applications.[7–9] For example, electron-donating or electron-withdrawing substituents are often introduced to produce a bathochromic shift in the emission spectra for bioimaging application via minimizing phototoxicity and the interfering effects of autofluorescence.[10]

For high-resolution imaging technologies in biomedical research, brightness and photostabilities of the fluorophores are critical, which are mainly achieved by employing nanotechnology.[11–15] Compared to other nanomaterials, fluorescent organic nanoparticles (FONs) have more advantages, including easy functionalization (fabrication), low toxicity, good biocompatibility, and low cost.[16–17] In contrast to the large number of studies reported for nanomaterials based on metals, polymers, quantum dots, and carbon dots,[16,18–22] nanoparticles (NPs) based on organic fluorophores have been rarely reported due to aggregation-caused quenching (ACQ) in the condensed phase, which has been observed for most conventional fluorophores possessing an aromatic system with high planarity.[23–24] For biomedical imaging, the development of FONs with high brightness in water is critical.[25–28] Up to now, FONs with a controlled morphology and tunable emission have been rarely developed.[29–31]

Recently, weak intermolecular interactions of organic fluorophores such as π-π stacking, van der Waals forces, and hydrogen bonds, which lead to different morphologies and optical properties in assembled states, have drawn considerable attention for the development of novel functional molecules in optoelectronics, biosensing, bioimaging, etc.[32–37] Therefore, substituent effects on the photophysical and morphological properties of NPs have been investigated with various scaffolds.[29,38–46] In addition, changing the position of

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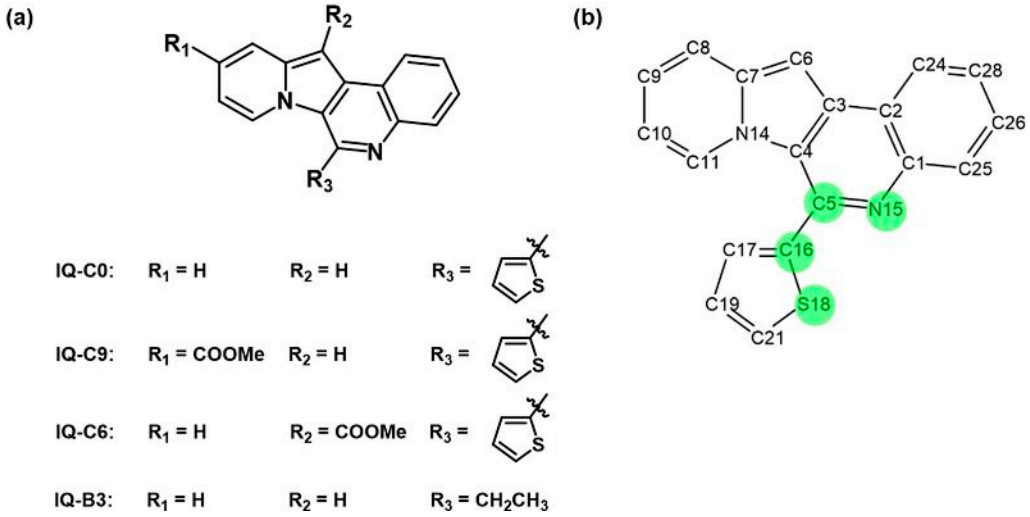


Figure 1. (a) Chemical structures of indolizino[3,2-*c*]quinoline derived fluorescent probes and (b) atom numbering and definition of the torsional angle (∠N15-C5-C16-S18 in green) between the quinoline and thiophene moieties in **IQ-C0**. Hydrogen atoms were removed for clarity in (b).

functional groups was suggested to adjust dipole-dipole directions for the self-assembly of luminogens.[47] Both the electronic contribution and position effect of substituents are regarded as crucial factors for tuning self-assembly, which generally requires extensive synthetic efforts along with the characterization of derivatives.

Our previous reports for 6-(thiophen-2-yl) indolizino[3,2-*c*] quinolines (IQs) suggested the importance of a methoxycarbonyl (MC) group attached to the C6 position of the IQ scaffolds in the presence of a thiophene group.[48–49] Herein, we explored optical properties by changing the position of the MC substituent, thereby modulating the HOMO-LUMO gap guided by the density functional theory (DFT) computational method (Figure 1a). We also explored the nature of the self-assembly of **IQ-C9** to understand the contribution of the substituents in the nanoparticles and in the solid states.

2. Results and Discussion

2.1. DFT-Calculation-Based Design to Select Positions for the Attachment of the MC Group

In previous reports from our group, we observed a significant contribution from the methoxycarbonyl (MC) group to the emission of the IQ scaffold, as shown in Table 1. This prompted us to investigate the contribution of the substituents in a systematic manner with the assistance of DFT calculations.

To characterize the photophysical properties of **IQ-C0** and **IQ-C6** and to understand the effect of the MC substituent on the blue shifts in the absorption and emission spectra of the **IQ-C6** relative to **IQ-C0**, the vertical excitation energies, oscillator strengths, and absorption wavelengths for several electronic vertical excitations for the two compounds in the DMSO solution were obtained via time-dependent (TD) DFT calculations. Table S1 shows that the main electronic absorptions and emissions with the largest oscillator strengths above the wavelength threshold of 350 nm for both compounds are directly related to $S_0 \rightarrow S_1$ and $S_1 \rightarrow S_0$ transitions, respectively, which are mainly characterized by the transition between HOMO and LUMO. The dependence of the frontier molecular orbital (MO) energy levels on the position of the MC group in the derivatives of **IQ-C0** is illustrated in Figure 2a and 2b (also see Table S2) where the frontier MO energy levels in Figure 2a were obtained from the optimized geometry of S_0 while those in Figure 2b were acquired from the optimized geometry of S_1. The Figures 2a and 2b reveal a lowering of the MO energy levels in the derivatives compared to **IQ-C0**, which is attributed to the presence of the electron withdrawing MC group in the derivatives. The MC substituent attached to **IQ-C0** draws electrons away from the indolizino[3,2-c]quinoline backbone towards itself, thereby reducing the repulsive interactions between electrons and decreasing the MO energy levels of the derivatives. The blue shifts in the absorption and emission spectra of **IQ-C6** and the spectral shifts in the other possible derivatives of **IQ-C0** can be examined more closely by introducing a HOMO-LUMO energy gap difference (ΔE_g), which is defined as [Eq. (1)]:

$$\Delta E_g = E_g^{IQ-Cx} - E_g^{IQ-C0} \quad (1)$$

Table 1. Experimentally measured and computationally calculated absorption and emission wavelengths [nm] of **IQ-C0** and **IQ-C6** in DMSO solutions. The calculated values are evaluated at the (TD) B3LYP/6-31+G(d,p) level of theory with the solvent effect using C-PCM.

Compound	Experimental		Calculated	
	Absorption	Emission	Absorption	Emission
IQ-C0	403	508	411.9	516.5
IQ-C6	366, 381	475	387.7	484.9

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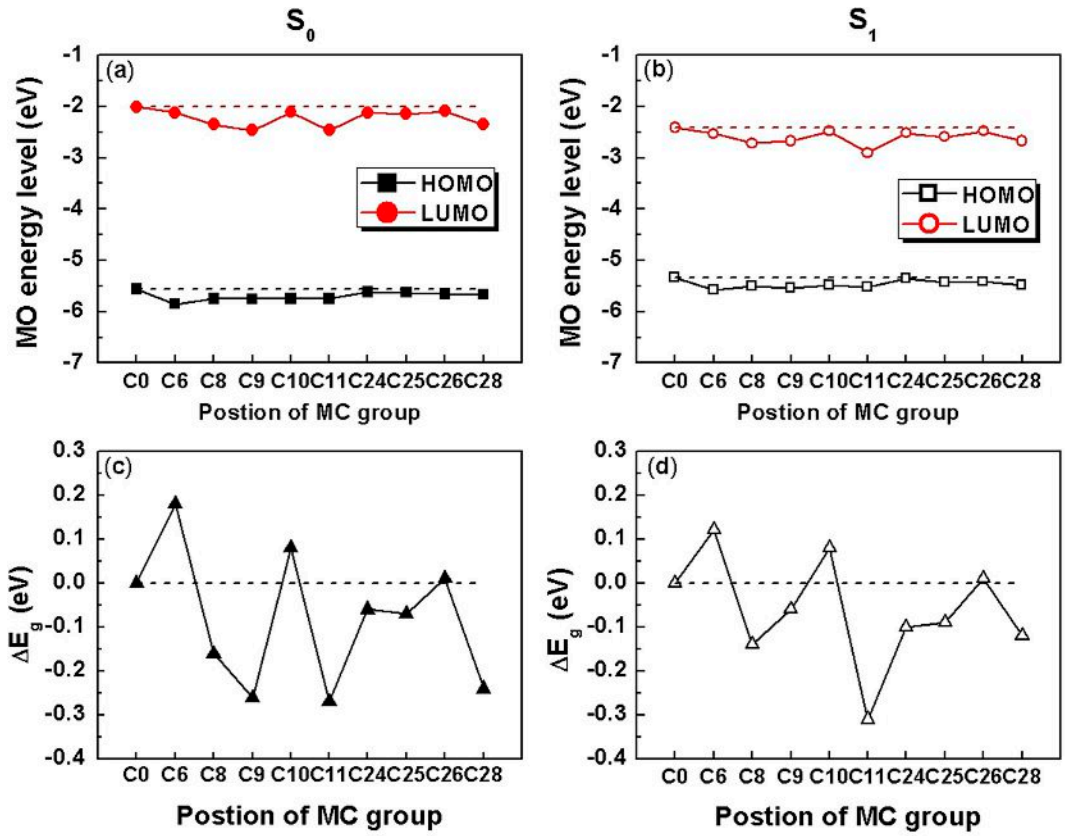


Figure 2. (a) and (b) Frontier orbital energy levels and (c) and (d) ΔE_g of **IQ-C0** and its possible derivatives in their optimized geometries of S_0 and S_1 in DMSO, respectively. In the labels for the x-axis, Cx (x = 0,6,8,···) represents the **IQ-C0** derivative with the MC group bonded to the position of the Cx atom of **IQ-C0** (see Figure 1b). The dashed horizontal lines are guides to the eye. DFT and TD-DFT calculations were performed at the same level of theory as in Table 1.

Here, E_g^{IQ-Cx} and E_g^{IQ-C0} are the HOMO-LUMO energy gaps of **IQ-Cx** and **IQ-C0**, respectively, and Cx represents an **IQ-C0** derivative with an MC substituent attached to the Cx position in **IQ-C0** (see Figure 1b). Consequently, a positive or a negative value of ΔE_g accounts for a blue or red shift in the spectra of the derivatives, respectively. Figure 2c and 2d clearly demonstrate the blue shifts in the **IQ-C6** spectra and also predict spectral shifts that will be observed in other derivatives of **IQ-C0**. ΔE_g evaluations for **IQ-C0** derivatives indeed indicate that the spectral shifts as well as the energy gaps of the **IQ-C0** derivatives can be modulated by altering the position of the MC substituent.

2.2. Synthesis of IQ-C9

Based on the calculation described in Figure 2c and 2d, attachment of the MC group at the C9, C11, and C28 positions appears to be reasonable for minimizing energy gaps of the **IQ-C0** derivatives. However, based on the steric hindrance at the C11 position with the thiophene ring and synthetic feasibility, the C9 position was selected for attachment of an MC group to lower the LUMO at the C9 position, thereby inducing a bathochromic shift for the new fluorophore (**IQ-C9**).

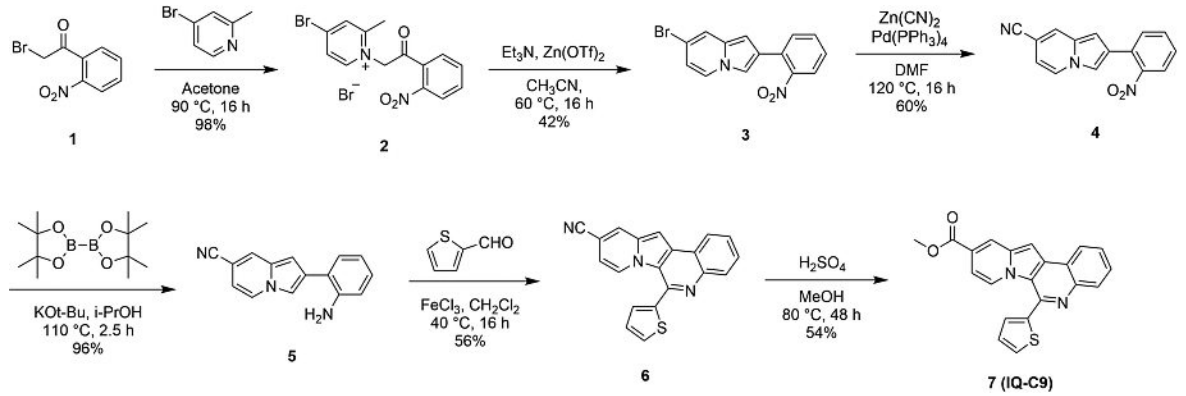
Scheme 1 showed the synthesis of **IQ-C9**. 7-bromo-2-(2-nitrophenyl)indolizine (**3**) was prepared from 2-bromo-2′-nitroacetophenone and 4-bromo-2-methylpyridine by base-mediated condensation with zinc salt. Substitution reaction of **3** with zinc cyanide in the presence of tetrakis(triphenylphosphine)palladium(0), followed by the reduction of a nitro group to an amino group using dioxaborolane[50] afforded 2-(2-aminophenyl)indolizine-7-carbonitrile (**5**). Further oxidative Pictet-Spengler cyclization and subsequent conversion of a nitrile group to an ester group yielded **IQ-C9** as a final product (overall yield: 7.2 %).

2.3. Optical Characterization and DFT Calculations

Figure 3 and Figure S1a depict the optical properties of **IQ-C9** in solution compared to **IQ-C6** and **IQ-C0**. In accordance with our rationale for calculating ΔE_g, a bathochromic shift (24 nm) was observed in the UV absorption spectra of **IQ-C9** after attachment of the MC to the C9 position of **IQ-C0. IQ-C9** exhibited a green emission in solution with a bathochromic shift of 34 nm from **IQ-C6** with blue emission. As summarized in Table 2, **IQ-C9** exhibited improved optical properties compared to previous compounds, including a high fluorescence quantum yield and high extinction coefficient in solution.

In Table S3, the calculated absorption and emission wavelengths obtained from the energy difference between the optimized geometries of S_0 and S_1 for **IQ-C0**, **IQ-C6** and **IQ-C9**

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Scheme 1. Synthesis of **IQ-C9**.

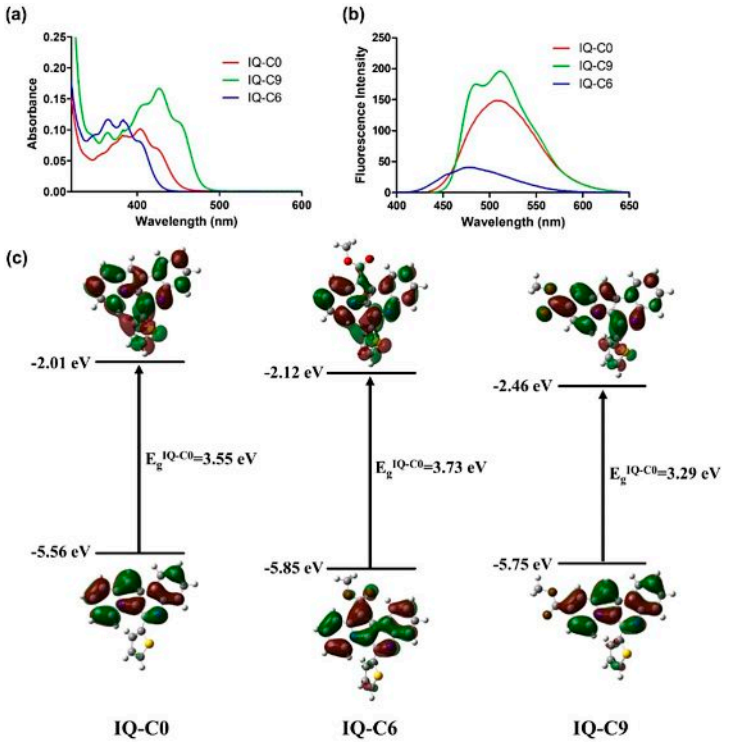


Figure 3. UV/Vis absorption (a; 10.0 μM) and emission (b; 2.0 μM) spectra of **IQ-C0**, **IQ-C9**, and **IQ-C6** measured in DMSO. Each compound was excited at the wavelength showing maximum absorbance: 403 nm (**IQ-C0**), 427 nm (**IQ-C9**), and 381 nm (**IQ-C6**). (c) MO energy levels, energy gaps (E_g) and isodensity surfaces for the HOMO and LUMO of **IQ-C0**, **IQ-C6**, and **IQ-C9** at the optimized geometry of S_0 in DMSO at the same level of theory in Table 1. Optimized geometries of the three compounds in S_0 and S_1 states are provided in Figure S2.

Table 2. Photophysical properties of **IQ-C0**, **IQ-C9**, and **IQ-C6** in the solution state.

Compound	λ_{abs}[a] (nm)	λ_{em}[b] (nm)				ε_{max} ($M^{-1}cm^{-1}$)	Φ_f[c]
		H_2O	EtOH	DMSO	THF		
IQ-C0	403	528.5	487.0	508.0	484.5	9848	0.20
IQ-C9	427	486.0, 517.5	475.0, 502.5	484.5, 511.0	474.5, 503.5	15047	0.41
IQ-C6	366, 381	493.0	470.0	475.0	470.0	12900, 12647	0.03

[a] Absorption maximum measured at the concentration of 10.0 μM in DMSO. [b] Emission maximum measured at the concentration of 2.0 μM.
[c] Determined in ethanol solution by using Coumarin 153 as a reference compound.

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in DMSO at the B3LYP/6-31 + G(d,p) level of theory have a good qualitative agreement with the measured absorption and fluorescence spectra in Table 2. For comparison, calculated absorption and emission wavelengths using other functionals are also presented. The spectra measured for **IQ-C9** in DMSO in Figure 3a and 3b indeed validate the ΔE_g calculations in Figure 2c and 2d which predict the red-shifts of the absorption and emission peaks of **IQ-C9** with respect to the spectra for **IQ-C0**. As is the case for **IQ-C0** and **IQ-C6**, the main electronic absorptions and emissions with the largest oscillator strength above the wavelength threshold of 350 nm for **IQ-C9** are directly related to $S_0 \rightarrow S_1$ transitions, which are also attributed to the HOMO→LUMO transitions (Table S4). For a closer examination of $S_0 \rightarrow S_1$ excitations in the S_0 optimized geometries for the three compounds, the energy levels and isodensity surfaces for the frontier MOs are presented in Figure 3c. The order of the HOMO - LUMO energy gap for the three compounds is $E_g^{IQ-C6} > E_g^{IQ-C0} > E_g^{IQ-C9}$ as expected because the frontier MOs make the largest contribution to the absorption and emission processes. The electron-withdrawing effect of the MC group attached to **IQ-C6** and **IQ-C9** is also clearly observed in Figure 3c. The electron densities of the thiophene ring in Figure 3c are shown to be much larger in the LUMO than in the HOMO for the three compounds, indicating that a substantial charge transfer from the indolizino[3,2-*c*] quinoline backbone to the thiophene ring takes place upon the $S_0 \rightarrow S_1$ excitation. The intramolecular charge transfer (ICT) character in the $S_0 \rightarrow S_1$ transitions was further confirmed using natural transition orbital (NTO) analysis, showing that the highest occupied transition orbital (HOTO, hole) mainly exists in the indolizino[3,2-*c*]quinoline moiety and that a large part of the lowest unoccupied transition orbital (LUTO, particle) resides in the thiophene ring (Figure S3).

The origin of the spectral shifts observed in the absorption and emission spectra of **IQ-C6** and **IQ-C9** relative to **IQ-C0** was assessed by calculating the fractional atomic orbital contribution (FAOC) coefficient, f_{xn} of an atom x to a specific MO n such as the HOMO or LUMO, which is given as [Eq. (2)]:

$$f_{xn} = \sum_{a \in x} \sum_{j} c_{an} c_{jn} S_{aj} \tag{2}$$

where a is a basis function or an atomic orbital (AO) which belongs to the atom x, j runs over all of the basis functions, c_{an} is the coefficient of the basis function a in an MO n, and S_{aj} is the overlap integral between the two basis functions, a and j.[51] Consequently, $\sum_x f_{xn} = 1$, where x runs over all of the atoms in a molecule. Because the HOMO and LUMO are mainly involved in the $S_0 \rightarrow S_1$ excitation and the MC group is attached to a carbon atom at the C6 or C9 position in **IQ-C6** and **IQ-C9**, respectively, it is imperative that the contribution of those atoms to the HOMO and LUMO should be taken into account. In Table 3, the contributions of the C6 and C9 atomic orbitals to the HOMO and LUMO of **IQ-C0** at the optimized geometry of S_0 are represented using the FAOC coefficients.

It is worth noting that the FAOC coefficient at the optimized geometry of S_0 shows a weak dependence on the functional while the HOMO and LUMO energy considerably varies depending on the functional. Consequently, further FAOC calculations were performed based on the B3LYP functional that is one of the most popular functionals. As mentioned above, attachment of the electron-withdrawing MC group to **IQ-C0** withdraws the electron density and lowers the HOMO and LUMO energy levels. Therefore, the energy lowering effect of the MC group on an MO energy level will be enhanced when a larger portion of the electron density (or a greater FAOC coefficient) in an MO resides at an atomic position where the MC group is positioned. For instance, in the case of C6 in **IQ-C0**, a much greater FAOC coefficient for the HOMO than that for the LUMO indicates that the HOMO takes a larger portion of the electron density at C6 than the LUMO does at the same atom. As a result, in **IQ-C6** where the MC group binds to the C6 position in **IQ-C0**, the HOMO energy level is lowered by a value of 0.29 eV much greater than that of the LUMO, which is reduced by only 0.11 eV as compared with **IQ-C0** (Figure 3c), demonstrating a stronger influence of the MC group on the HOMO than the LUMO at the C6 position. On the other hand, a greater FAOC coefficient of C9 for the LUMO than for the HOMO in Table 3 implies that a bigger portion of the electron density at C9 is taken by the LUMO than by the HOMO. Consequently, the LUMO energy level of **IQ-C9** is lowered by 0.45 eV, while that of the HOMO is reduced by only 0.19 eV in Figure 3c. The same reasoning can be applied to the emission spectra for the three compounds using the energy levels of the frontier MOs and the FAOC coefficients calculated at the optimized geometries of S_1 (provided in Figure S4 and Table S5).

The relationship between the FAOC coefficient and the change in the frontier MO energy levels was investigated more thoroughly by evaluating the FAOC values for all the carbon

Table 3. Frontier orbital energy levels and fractional atomic orbital contribution (FAOC) coefficients (f_x) of the C6 and C9 atoms for the HOMO and LUMO at the S_0 optimized geometry of **IQ-C0** using the six different density functionals with the 6–31 + G(d,p) basis set in DMSO.

	B3LYP			M06-2X			PBE0			CAM-B3LYP			ωB97XD			B2PLYPD		
	E (eV)	f_{C6} (%)	f_{C9} (%)	E (eV)	f_{C6} (%)	f_{C9} (%)	E (eV)	f_{C6} (%)	f_{C9} (%)	E (eV)	f_{C6} (%)	f_{C9} (%)	E (eV)	f_{C6} (%)	f_{C9} (%)	E (eV)	f_{C6} (%)	f_{C9} (%)
LUMO	-2.01	0.8	9.9	-1.19	1.0	9.0	-1.90	0.8	9.5	-0.82	0.9	9.6	-0.23	0.7	9.9	-0.69	1.0	9.8
HOMO	-5.56	25.8	7.1	-6.74	25.7	7.1	-5.79	25.7	7.0	-6.80	25.8	7.1	-7.37	26.0	7.0	-6.25	25.0	7.2

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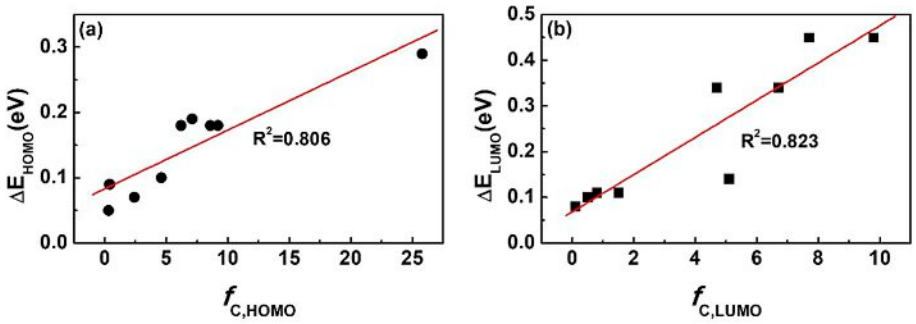


Figure 4. Relationship between the FAOC coefficients (f_x) of the carbon atoms in **IQ-C0** and the lowering of the MO energy levels (ΔE_{MO}) in **IQ-C0** derivatives for (a) the HOMO and (b) the LUMO. The geometry optimizations for all the derivatives at S_0 in DMSO were performed using the B3LYP/6-31+G(d,p) level of theory.

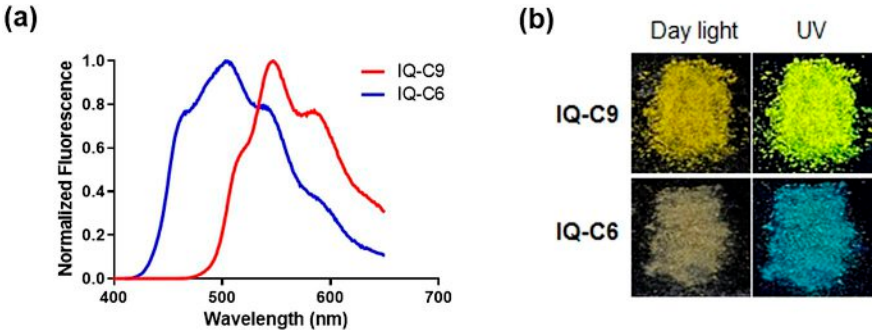


Figure 5. Comparison of photoluminescence properties of **IQ-C9** (red) and **IQ-C6** (blue) in the solid state. (a) Normalized solid emission spectra. Each compound was excited at the wavelength showing maximum absorbance: 368 nm (**IQ-C9**) and 390 nm (**IQ-C6**). (b) Photographs of the powder sample under visible light (left) and UV light (right, λ_{ex} = 365 nm), respectively.

atoms in **IQ-C0** to MC-group attachment and by comparing the FAOC values with the frontier MO energies of **IQ-C0** derivatives that are produced when an MC group is attached to such carbon atoms (Table S6). Here, the lowering of the frontier MO energy levels for an **IQ-C0** derivative, ΔE_{MO}, is defined as [Eq. (3)]:

$$\Delta E_{MO} = E_{MO}^{IQ-C0} - E_{MO}^{IQ-Cx} \quad (3)$$

where E_{MO}^{IQ-C0} and E_{MO}^{IQ-Cx} are the frontier MO energy levels of **IQ-C0** and a derivative, respectively. Figure 4 represents good linear relations between the FAOC coefficient and the lowering of the frontier MO energy levels of the **IQ-C0** derivatives at S_0 in DMSO despite some deviations in the LUMO. It is noted that in the case of the **IQ-C0** derivatives, the lowering of the LUMO energy levels shows a greater dependence on the FAOC coefficient than that of the HOMO energy levels, and as a result it is difficult to tell whether a blue or red shift occurs when the FAOC coefficient for the HOMO is slightly larger than that for the LUMO. Nevertheless, comparisons with the experimental results verify the validity of the FAOC analysis, which can be employed as a useful computational tool for prediction and design of a novel fluorescent compound.

2.4. Solid-State Emission and FON Characteristics of IQ-C9

Figure 5 depicts the emission spectra of **IQ-C6** and **IQ-C9** in the solid state. The maximum excitation of **IQ-C9** in the solid state occurs at 368 nm (Figure S5). The fluorescence quantum yield (QY) of **IQ-C9** in the solid state was 18%, exhibiting higher fluorescence than **IQ-C6** both in solution and in the solid state (Table 4). To obtain insight into the excited states of **IQ-C9**, fluorescence lifetimes in solid and solution states were measured using the TCSPC method. Figure S1b and S1c display the fluorescence decay profiles measured for **IQ-C9** with an average lifetime of 3.2 ns in solution and 4.0 ns in the solid state, indicating that the radiative pathway was enhanced in the solid state.

The observed fluorescence enhancement of **IQ-C9** in solid state was explored computationally by comparing HOTO and LUTO isodensity surfaces for the $S_0 \rightarrow S_1$ transition in the solid state with those in THF solution. To model the optimized ground (S_0) state geometry of **IQ-C9** in the solid state, the initial geometry of **IQ-C9** was taken from the crystal structure in Figure 7a and was partially optimized in the gas phase at the B3LYP/6-31+G(d,p) level of theory with the dihedral angle (defined in Figure 1b) fixed at its initial value of −98.9° in the crystal structure. On the other hand, the S_0 state geometry of **IQ-C9** in THF solution was fully optimized without any restraints at the same level of theory. The NTO analysis in Figure S6

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Table 4. Photophysical properties of **IQ-C9** and **IQ-C6** in the solid state. [a] Determined by using an integrating sphere. [b] Determined by following equation; $\Phi_f = k_f/(k_f + k_{nr})$ and $< \tau_f > = 1/(k_f + k_{nr})$

Compound	λ_{abs} (nm)	λ_{em} (nm)	Φ_f[a]	$<\tau_f>$ (ns)	k_f[b] (ns^{-1})	k_{nr}[b] (ns^{-1})
IQ-C9	368	547	0.18	4.0	0.044	0.202
IQ-C6	390	505	0.05	5.9	0.008	0.160

[a] Determined by using an integrating sphere. [b] Determined by following equation; $\Phi_f = k_f/(k_f + k_{nr})$ and $< \tau_f > = 1/(k_f + k_{nr})$.

illustrates that while a substantial charge transfer from the indolizino[3,2-c]quinoline moiety to the thiophene ring is observed for the $S_0 \rightarrow S_1$ excitation in THF solution with a smaller dihedral angle of $-46.6°$ (Figure S6a), the $S_0 \rightarrow S_1$ transition in the solid state is mainly characterized by the local excitation (LE) inside the indolizino[3,2-c]quinoline backbone due to the restriction in the dihedral angle (Figure S6b), which can open a new channel for the radiative decay of **IQ-C9** in the solid state.

The fluorescence properties of **IQ-C9** were investigated in both organic solution (THF) and water by changing the water volume fraction (f_w) (Figure 6a). An increase in the water volume fraction in THF solution of up to 80% resulted in a dramatic enhancement of the emission; red-shifted emission maxima (from 502 nm to 570 nm) with decreased fluorescence intensity were observed for a water volume fraction >80%. The decrease in emission intensity at >80% water is due to the formation of large amorphous aggregates (~ μm). When sonicated during the addition of a stock solution of **IQ-C9** in THF to water, the aqueous sample forms nanoparticles, with a good colloidal stability of the particles reflected by the zeta potential value of −33.7 mV (Figure S7b). These results suggested the potential of **IQ-C9** for use as FONs in aqueous suspension for biomedical imaging applications.

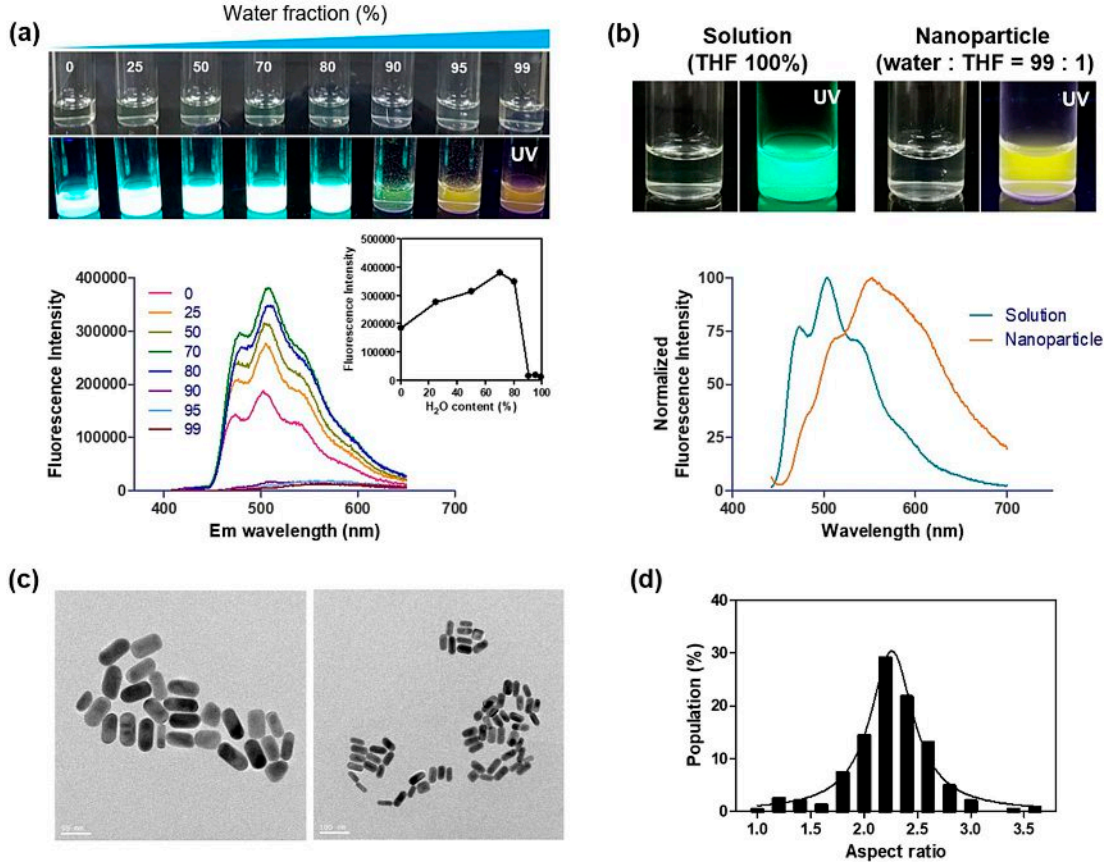


Figure 6. (a) Photographs of 10 μM **IQ-C9** in a water/THF mixture (0–99%) under visible light (up) and UV light (down, $\lambda_{ex} = 312$ nm), and the fluorescence spectra measured with excitation at 427 nm; inset: plots showing the fluorescence intensity of **IQ-C9** at 510 nm. (b) Photographs of 10 μM **IQ-C9** solution (in THF, left) and 10 μM **IQ-C9** nanoparticles (water : THF = 99:1, v/v, right). Normalized emission spectra of both samples are shown below ($\lambda_{ex} = 427$ nm). (c) TEM images of 1 μM **IQ-C9** nanoparticles (water : DMSO = 90:10, v/v). The scale bars represent 50 nm (left) and 100 nm (right). (d) Aspect ratio distributions of (c). The average value and standard deviation is 2.3 and 0.25, respectively.

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The emission maximum at 502 nm in 100% THF solution (green) was shifted to 550 nm (yellow) in NP dispersed in 99% water solution (Figure 6b). The fluorescent quantum yield of the dispersed NP (water : THF = 99:1) was measured to be 4.6%. As depicted in Figure S1a, the effect of solvent polarity in **IQ-C9** appears to be negligible, indicating that the observed bathochromic shift was not a result of a solvatochromic effect. TEM images (Figure 6c and Figure S8) demonstrated that **IQ-C9** forms well-defined nanoparticle structures (dimensions: 20× 50 nm, aspect ratio: 2.3) in aqueous solution. Analysis of the aspect ratio shown in the histogram (Figure 6d) revealed that **IQ-C9** forms nanoparticles with uniform shapes and sizes. DLS data suggested that the average size of the nanoparticles is 129.7 ± 27.6 nm (Figure S7a). The span value, which indicates the width of the particle size distribution, was 0.559. The discrepancy in the nanoparticle size observed from TEM may be due to the nonspherical shape of the particle. The size of the particles appears to be dependent on the water volume fraction, sample concentration, and preparation method. The stability of the NP was measured to be high in 1% DMSO aqueous solution (Figure S7d).

2.5. X-Ray Diffraction Data of IQ-C9

To understand the intermolecular interactions that contribute to solid state emission, the geometry and packing arrangements in the X-ray crystal structure were analysed (Figure 7). A slipped stack (J-aggregate) with a slip angle of 46.7° was observed, minimizing both the conventional π - π stacking between IQs and the nonradiative decay that induces ACQ. The relatively high fluorescence quantum yield of **IQ-C9** in the solid state is in agreement with Kasha's molecular exciton model of the J-aggregate.[33,52]

Intermolecular and intramolecular interactions are shown by different coloured lines. Nonclassical hydrogen bonds involved with MC groups (intermolecular CH···O: 2.68 Å; intramolecular CH···O: 2.44 Å) and intermolecular S···S interactions (4.09 Å) and CH···S bonds (3.05 Å) with the thiophene ring comprise tight networks in the X-ray structure. Intramolecular hydrogen bonds between the H of indolizine and the S of the thiophene ring (CH···S: 2.79 Å) also contribute to the suppression of the rotation of the thiophene ring.

2.6. Bioimaging of IQ-C9 Using Confocal Microscopy

Figure 8 displays live cell images of MCF7 cells treated with **IQ-C9** using confocal microscopy. Cytotoxicities of 0.5, 1, 2, 5, 10 μM of **IQ-C9** were found to be negligible against MCF7 and HeLa cells (Figure S9). As shown in Figure 8a, sharp fluorescent puncta were observed in green and yellow emission. Colocalization experiments with LysoTracker and red fluorescent protein-tagged LC3 proteins transfected with pmRFP-LC3, a marker for autophagosomes, did not show overlap with the **IQ-C9** puncta (Figure S10). Meanwhile, a Deep-Lipid Tracker for the lipid droplet was well colocalized with **IQ-C9** puncta, with Pearson coefficient and overlap coefficient values of 0.765 and 0.825, respectively (Figure 8b). Lipid droplets are highly dynamic organelles within cells, serving as storage for neutral lipids and regulating cellular lipid metabolism. Further studies of IQ derivatives for imaging cellular dynamics associated with lipid droplets are underway.

3. Conclusions

In this study, HOMO-LUMO energy gap analysis using the DFT method were employed to design and synthesize **IQ-C9** with a specific range of emission wavelengths. The photophysical properties of **IQ-C0**, **IQ-C6**, and **IQ-C9** and the origin of the spectral shifts of the absorption and emission peaks were characterized both experimentally and computationally. The theoretically predicted blue shifts for **IQ-C6** and red shifts for **IQ-C9** in the absorption and emission spectra compared to **IQ-C0** match well with the experimental trends. Optical properties

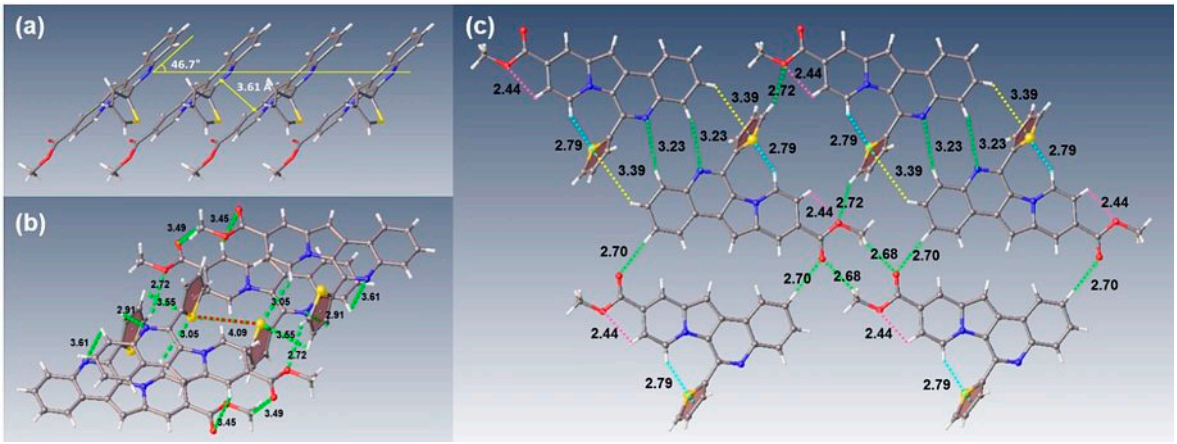


Figure 7. Crystal structure and stacking mode of **IQ-C9**. (a) Molecular packing diagram. (b,c) Packing structures including inter- and intramolecular interactions; cyan: intramolecular CH-π interaction, yellow: intermolecular CH-π interaction, pink: intramolecular hydrogen bonding, green: intermolecular hydrogen, red: intermolecular S–S interaction.

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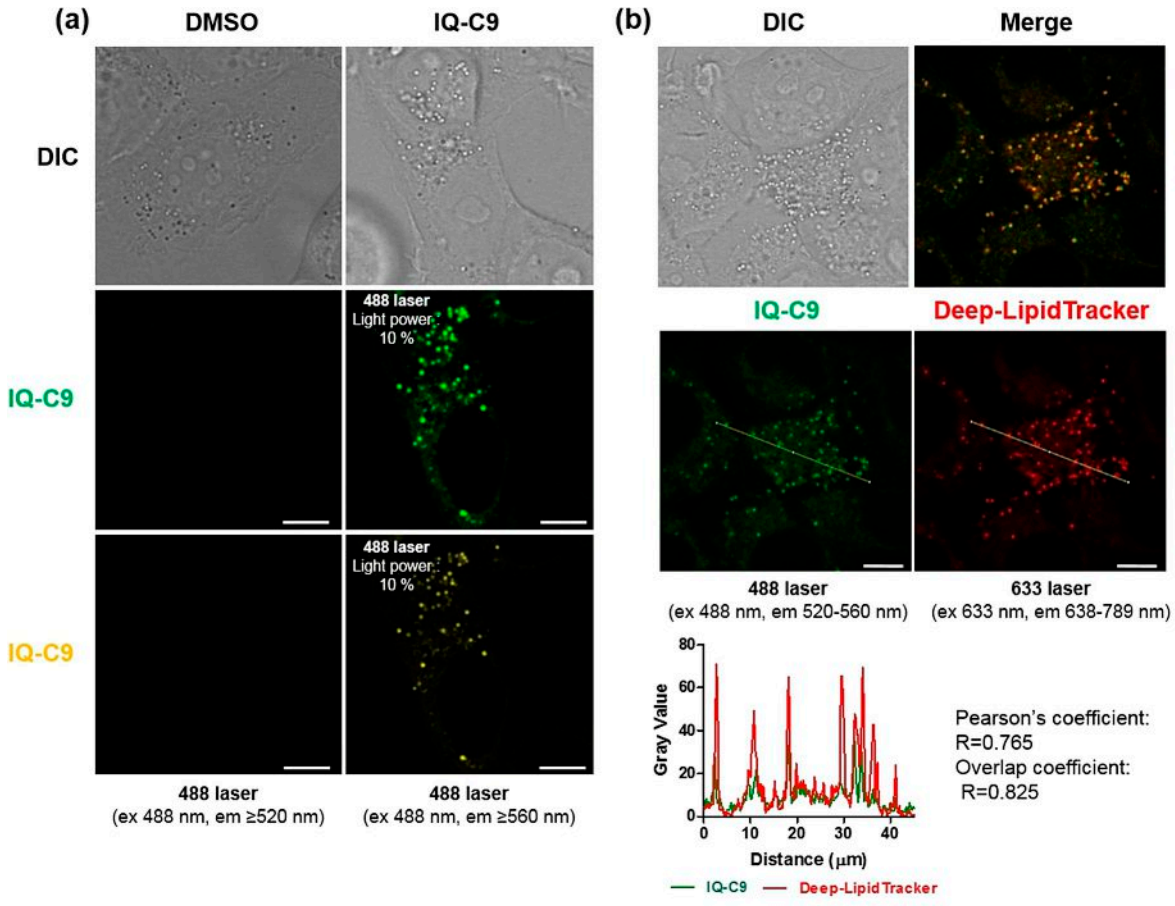


Figure 8. Confocal live-cell images of **IQ-C9** in MCF7 cells. (a) Images of MCF7 cells pretreated with 10 μM **IQ-C9** for 1 h. The green channel (Ex: 488 nm, Em: ≥520 nm) and the yellow channel (Ex: 488 nm, Em: ≥560 nm) arose from **IQ-C9** fluorescence. The scale bar represents 10 μm. (b) Images of MCF7 cells pretreated with 10 μM **IQ-C9** for 1 h, followed by treatment with Deep-LipidTracker (1:1000 dilution) for 30 min. The green channel (Ex: 488 nm, Em: 520–560 nm) arose from the **IQ-C9** fluorescence, and the red channel (Ex: 633 nm, Em: 638–789 nm) from Deep-LipidTracker fluorescence. Intensity profile of the region of interest (ROI) presented with a white line across the MCF7 cells. Pearson's coefficient: R = 0.765, Overlap coefficient: R = 0.825.

of the newly synthesized **IQ-C9** with the MC group attached to the C9 position were consistent with our DFT calculations, confirming the role of the MC group at the C9 position in lowering the LUMO energy level more than HOMO, thereby leading to a bathochromic shift via a decrease in the HOMO-LUMO gap. In addition, **IQ-C9** exhibited higher QYs in solution and in the solid state compared to **IQ-C6** with the MC group at the C6 position, and those improved QYs in both states lead to a dual-state emission (DSE) property of **IQ-C9**. Self-assembled structures of **IQ-C9** revealed the formation of well-defined **IQ-C9** nanoparticles. The characteristics of the fluorescent organic nanoparticles (FONs) based on **IQ-C9** enabled us to investigate the imaging properties using fluorescent confocal microscopy, revealing that **IQ-C9** is specifically localized in lipid droplets in live MCF7 cells. Our study demonstrates that FAOC coefficient analysis can be used as a computational tool to predict the spectral behaviour of fluorophores and to design a novel fluorescent biosensor with a specific range of emission wavelengths.

Experimental Section

For Experimental details, please see the Supporting Information.

Supplementary crystallographic data of **IQ-C9** in this paper can be downloaded from the Cambridge Crystallographic Data Centre (Cambridge CB2 1EZ, UK) via www.ccdc.cam.ac.uk/data_request/cif. (Deposition number: CCDC-1949818).

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Conflict of Interest

The authors declare no conflict of interest.

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